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Heart Disease Prediction in Indonesia Using Machine Learning Techniques for Multiclass Classification

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***Abstract*—This research concentrates on utilizing multiclass classification methodology for heart disease prediction. Specifically, this paper explores the utilization of Support Vector Machine (SVM) and Multi-Layer Perceptron (MLP) classifiers to classify various cardiac conditions. This research is used to get the best results in predicting heart disease that has a classification of more than two categories using machine learning methods. The main goal is to contribute insights to medical diagnostics by explaining the effectiveness of SVM and MLP in improving the accuracy of heart disease prediction models in various classes using several factors that influence heart disease. The data used in this research has missing values, therefore in this research, several methods were used to overcome missing values. Determining the best model is done by comparing precision, recall, and accuracy values. The best model in this research is the SVM model because it has the highest accuracy, precision, and recall. It is hoped that this research can provide knowledge to optimize its application in the identification and classification of heart disease to make it more efficient.**

Keywords—heart disease, multiclass, SVM, MLP Classifier, classification

I. INTRODUCTION

Heart disease is still one of the biggest global health issues and causes of mortality. In low- and middle-income nations, such as Indonesia, the most populous nation in Southeast Asia and home to the fourth biggest population in the world, the number keeps rising. Cardiovascular disease (CVD) is one of the leading causes of premature mortality in the area, encompassing stroke and coronary heart disease (CHD). Growing life expectancy, urbanization, demographic shifts, epidemiological transition, and the high prevalence of CVD risk factors are all contributing reasons to the rise in CVD incidence and risk factors in emerging nations like Indonesia [1]. Heart failure-related fatalities are becoming more commonplace daily. 32 percent of all fatalities worldwide in 2019 were attributable to cardiovascular illnesses, which claimed 17.9 million lives, according to a WHO report. Heart attacks and strokes accounted for 85% of these fatalities. The cause of this is that individuals do not know why they have coronary heart disease [2].

Coronary heart disease can be caused by a variety of causes, including high blood pressure, smoking, diabetes, high cholesterol, and other conditions. Smoking is growing more and more frequent among young people and the elderly. Smoking raises blood stress and constricts the coronary heart's arteries, which results in irregular heartbeats. Numerous problems are caused by high blood pressure. It also causes the lower-left coronary heart chamber to thicken, increasing the risk of coronary heart failure. These two effects drive the coronary heart to pump blood to the body harder [3]. To classify diseases, healthcare analytics which is now frequently using restricted machine learning applications or conventional statistical approaches. But these methods might not be as accurate and effective as they could be, which would need researching more sophisticated methods [4].

The research of a human disease prediction model based on the Random Forest algorithm has been proposed [5]. The model is expected to be able to predict possible diseases based on the symptoms entered by the user. The model itself used Random Forest, Long-Short Term Memory (LSTM), and Support Vector Machine (SVM) algorithm and a dataset from Kaggle in order to train the data. The result of this proposed model managed to produce a 97% accuracy in predicting possible diseases compared to older models [5]. The fields of disease populations, disease status, immune response, and health emergencies have all advanced significantly as a result of recent advances in artificial intelligence (AI) and machine learning (ML) technologies. Though there is still doubt about the usefulness of ML-based techniques in practice and how to interpret their findings, their use in healthcare settings is growing quickly [6].

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In the pursuit of enhancing the accuracy of heart disease prediction, this study explores the implementation of multiclass classification techniques. As heart diseases manifest in various forms, accurate classification is imperative for effective medical decision-making. Traditional binary classification models often fall short of addressing the nuanced complexity of multiclass scenarios. Hence, the focus on multiclass classification aims to overcome these limitations and provide a more comprehensive framework for predicting a spectrum of heart-related ailments. This paper investigates the application of two powerful machine learning classifiers: Support Vector Machine (SVM) and Multi-Layer Perceptron (MLP). The objective is to find the best method to systematically categorize diverse types of heart conditions, enabling a nuanced understanding of the intricate patterns and relationships within heart disease data.

In this research, classification modeling will be carried out on heart disease patients using a multiclass classification approach, considering that heart disease conditions can be categorized into several levels of severity, such as normal, mild (Level 1), moderate (Level 2), and severe (Level 3). Two supervised learning methods, namely Support Vector Machine (SVM) and Multilayer Perceptron (MLP), were chosen because each has advantages in handling complex and nonlinear data classification. The analysis will focus on comparing the performance of the two algorithms. Evaluation of model performance will be carried out using various metrics that are suitable for multiclass classification, such as accuracy, precision, recall, and F1-score. Thus, this research is expected to contribute to the development of decision support systems in the health sector, especially in data-based early diagnosis of heart disease.

II. RELATED WORK

Multiclass classification for heart disease was also conducted by Hasan, Jasim and Hashim. The study used data from Ibn Al-Bitar Hospital Cardia Surgery and Baghdad Medical City, the results showed that prediction using the MLP method was better than SVM [7]. Heart disease research using data from the Cleveland database was also conducted for binary classification and multiclass classification. The results showed differences between the two cases, for binary classification it is better to use SVM, while for multiclass classification it is better to use MLP [8]. Boukhatem, Youssef, and Nassif compared the MLP, SVM, RF, and NB methods for heart disease classification. The results showed that the SVM model was better than other models [9]. While Laveniya and Harun used binary classification for heart disease and the results showed that the MLP method was better than various other machine learning methods [10].

III. SUPPORT VECTOR MACHINE

One of the supervised learning methods that can be used for classification and regression is the Support Vector Machine (SVM). The concept of SVM is to maximize the geometric margin to minimize the empirical classification error. Thus, another term for the SVM is the Maximum Margin Classifier. The basis of SVM is structural risk minimization (SRM). SVM creates a maximal separating hyperplane by mapping the input vector to a higher dimensional space. Two parallel hyperplanes are constructed on either side of the hyperplane that splits the data. The separating hyperplane is the one that maximizes the distance between the two parallel hyperplanes. The more margin between these parallel hyperplanes, the better the generalization error of the classifier [11].

SVM has been widely implemented in Health Data Analysis. One of them is applied to the automatic classification of microarray gene expression profiles. Theoretically, SVM can examine gene expression profiles obtained from tumor samples [12].

Class classification problems were primarily supported by vector machines, however as computer, network, and database technologies advanced quickly and became more sophisticated, making it possible to classify and manage vast amounts of information, class classification problems were unable to keep up with societal demands. The multi-class classification issue is extended to use this strategy. The two categories of multi-class classification problems in the current treatment methods are as follows: To solve multi-class classification issues, first construct a sequence of second-class classification problems in some manner. The final class classification problem is based on two multi-class SVM classification methods, in essence. Secondly, create a single multi-class support vector machine (SVM) and use it to solve the multi-class classification problem [13]. Equation (1) below is used to process each class simultaneously in multiclass classification. Thereafter, utilize equation (2) to make a conclusion [14].

$$min_{nh_1 b_1 sv} \frac{1}{2} \sum_{y=1}^{q} nh_y^p nh_y + c \sum_{i=1}^{r} \sum_{y \neq r_1} sv_{i_1 y} \quad (1)$$

$$decn = max_y \left(w_v^p \beta(x_i) + b_x \right) \quad (2)$$

Where y is the number of classes, r is the number of samples, sv is the slack variable, and b is the bias. Every pair of classes receives the application of every classifier. The class with the highest number of votes determines the final classification. There are many types of kernel functions in SVM. The kernel function is used to take input data and shape it into an appropriate form. The following are frequently used kernel functions.

A. Linear Function

A linear kernel, also referred to as a soft margin, may accept one or more data misclassifications while still attempting to identify a straight-line hyperplane. Even though the linear kernel can tolerate some errors, it nevertheless searches for a line that maximizes margins and minimizes misclassification. The degree of misclassification tolerance set has a significant impact on the hyperplane's accuracy. The tolerance in Sklearn is referred to as C. The margin is smaller and misclassification tolerance is lower with greater values of C. Equation (3) is a linear kernel function.

$$K(x, z) = x.z + C \quad (3)$$

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B. *Polynomial Function*

A polynomial kernel is required when data cannot be divided into distinct categories by lines. Nonlinear decision boundaries may be produced using polynomial kernels. A polynomial kernel generates a new feature by applying a polynomial combination of the existing features. The polynomial kernel function is written in Equation (4).

$$K(x,z) = (\gamma x.z + C)^d, d > 0 \quad (4)$$

C. *Radial Basis Function*

A Radial basis function (RBF) kernels are used on highly unevenly distributed data. Two parameters—C and gamma—for highly uneven data require a radial basis function (RBF) kernel. Two parameters—C and gamma—must be taken into account when training a dataset with the RBF kernel. The training data classification error decreases when the C parameter is larger, because it indicates the magnitude of errors that must be avoided in the classification process. The gamma parameter is the magnitude of the impact of one training data sample. As a result, the distance from the data points to be calculated increases as the gamma value decreases [15].

$$K(x,z) = exp\left(-\gamma||x-z||^2\right), \gamma > 0 \quad (5)$$

D. *Sigmoid Function*

A Sigmoid function is applied to non-linear classification issues. The sigmoid kernel is used to transform the input data into a higher-dimensional space [16]. The following is the sigmoid function equation.

$$K(x_i, y_j) = tanh(\gamma x_i, x_j + r) \quad (6)$$

Based on the kernel function above. Fig. 1 is an example of a kernel function implemented on Iris data [17].

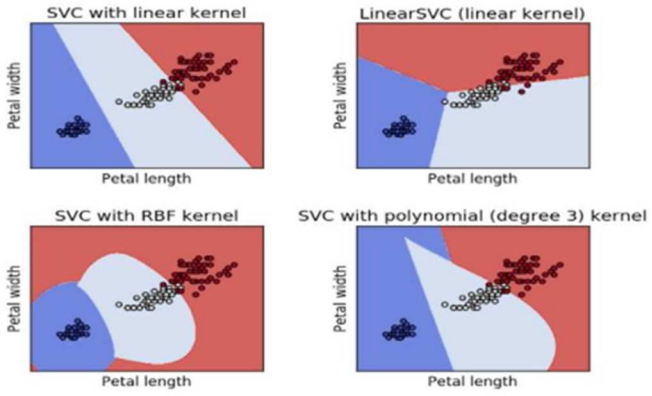


Fig. 1. Kernel Functions in SVM

IV. MULTILAYER PERCEPTRON (MLP) CLASSIFIER

One of the main focuses of neural network research is choosing the appropriate neural network architecture to address a certain issue. One or more hidden layers make up the multilayer perceptron (MLP). The convergence of MLP training methods is significantly influenced by the number of neurons in the hidden layers and the initial weights [18]. The Feed Forward Neural Network (FFNN) architecture is one type of artificial neural network. The FFNN architecture is used by the MLP Classifier. This classifier learns to categorize with a multilayer perceptron using backpropagation. The input layer, hidden layer, and output layer are the three layers that make up the MLP Classifier [19]. Both the input layer and the hidden layer have several nodes. The number of nodes in the input layer is an independent variable. On the other hand, the hidden layer could have one or more layers and nodes [20]. Processing from the input layer to the hidden layer and then to the output layer is called forward propagation. Then the activation function value is propagated to the next layer. The next layer can be another hidden layer or an output layer. MLP is trained from training data through processing which is called backpropagation. This process can be described as a way to improve errors [21]. Fig. 2 shows the architecture of FFNN. Meanwhile, the general model of MLP is written in equation (7) [22].

$$y = \varphi(w^T x + b) \quad (7)$$

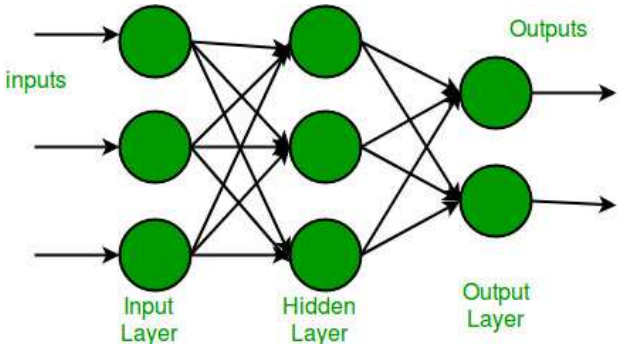


Fig. 2. The Architecture of Neural Network

An artificial neuron's activation range is determined by its activation function. This is applied to the total of the neuron's weighted input data. Non-linearity is a characteristic of an activation function. The linear products between the weights and the input values would be the only operations involved in calculating the output of a multilayer perceptron if an activation function wasn't used [23].

V. DATA AND METHODS

The data used in this paper was obtained from one of the hospitals in Indonesia. The samples in this research were patients at the hospital, whether patients suffered from heart disease or not. The number of samples in this research was 9756. However, the data in this research contained many missing values. There are dependent and independent variables used in this research. Table 1 shows the independent variables and Table 2 shows the levels of dependent variable.

There are two methods that can be used to carry out multiclass classification, namely SVM and MLP [24]. The stages of this research are as follows. First, perform Pre-Processing on the data to overcome missing values. In this research, there are two ways to do this, namely filling in empty data with median values and deleting empty data. Second, carry out analysis using SVM on data that has no missing values using four kernel functions. Third, carry out analysis using MLP on data that has no missing values using three activation functions. Fourth, compare the SVM and MLP models to get the best model using accuracy, precision and recall values.

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TABLE I. INDEPENDENT VARIABLES

Description	Code
Age of the patient	Age
Glucose	GLU
Cholesterol	CHOL
2-hour glucose is a test to determine blood sugar levels after 2 hours of eating.	GLU2
Triglycerides	TRIG
High-density lipoprotein (HDL) cholesterol	HDL
Urea	UREA
Creatine kinase	CREA
Urinalysis	UA
The total protein	TP
Albumin	ALB
Globulins	GLOB
Total bilirubin	TBIL
Bilirubin	DBIL
Serum glutamic oxaloacetic transaminase	SGOT
Gender of the patient (Female or Male)	Gender

TABLE II. LEVELS OF DEPENDENT VARIABLE

Status Levels	Description
Normal	Do not suffer from heart disease
Level 1	Level I heart disease sufferers without any activity limitations. Daily physical activity does not cause fatigue, palpitations, shortness of breath, and chest pain.
Level 2	Level II heart disease sufferers with slight limitations during moderate-intensity activities (such as climbing stairs). Complaints disappear with rest.
Level 3	Level III heart disease sufferers with clearly visible limitations during light-intensity activities (such as walking a few meters). Complaints will decrease with rest.

VI. DATA PRE-PROCESSING

Before carrying out classification analysis on heart disease patient data, pre-processing is first carried out. This is because the data in this research has many missing values. Then handling missing data must be done first. There are several variables that have missing values. In this research, variables that have missing values are imputed using the Median value for the numerical variables. This is because there are outliers in these variables, then the imputation uses the Median value. Numerical variables that were imputed using median values included Total Protein, Albumin, Globulins, Total Bilirubin, Bilirubin, and Serum Glutamic Oxaloacetic Transaminase. Meanwhile, categorical variables that have missing values are imputed using the category that has the most frequency. The imputed categorical variables include Glucose, Cholesterol, Triglycerides, High-density lipoprotein (HDL) cholesterol, and Urea. After that, the data is divided into training and testing data with a ratio of 80% training data and 20% testing data.

VII. RESULTS AND DISCUSSION

Data visualization on the Heart Disease Status variable is presented in Figure 3. This analysis is used to determine the frequency of each category in the dependent variable. Based on Figure 3, it can be concluded that the Normal category is the most dominant category, indicating that the majority of individuals do not have heart disease. While, Level 2 and Level 3, as severe categories, have the smallest number, indicating that high severity only occurs in a small number of individuals.

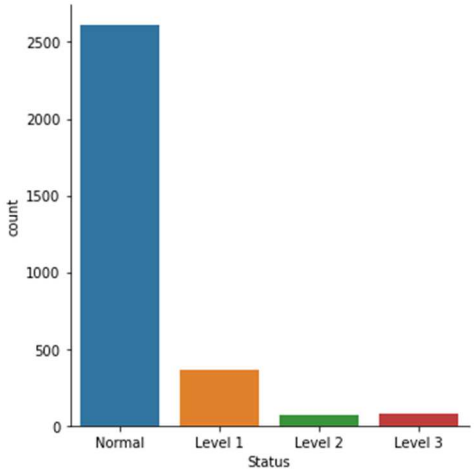


Fig. 3. The Bar Chart for the Heart Disease Variable

Figure 4 is a boxplot of total protein at each level of heart disease. Based on Figure 4, it can be concluded that there are outliers at each level of heart disease. Outliers are found at values that are smaller than the lower limit or values that are greater than the upper limit. This shows that the total protein in each category of heart disease varies greatly. The same thing happens to all numeric variables in this research.

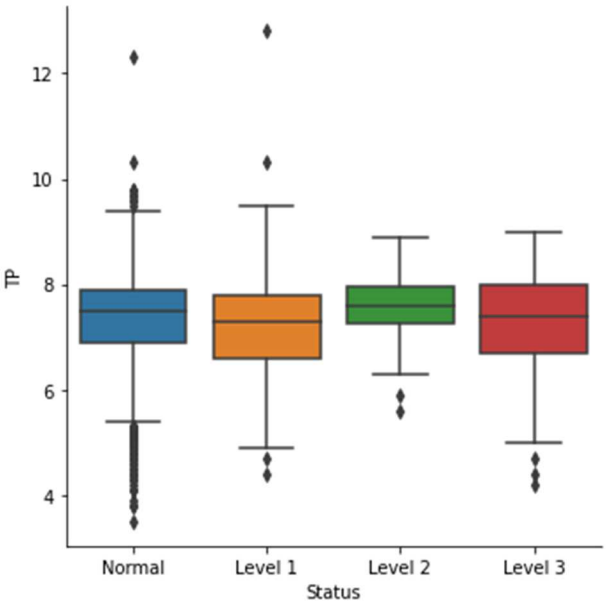


Fig. 4. The Boxplot of Total Protein for Each Heart Disease Class

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Furthermore, in this reseach two methods are used in data modeling. After imputation is carried out on the missing values, modeling is then carried out to predict the level of heart disease in patients. First, using the SVM method. In this method, data is trained using several kernel functions and to obtain the best model in the SVM method, hyperparameter tuning is carried out on the SVM model. Hyperparameter tuning is done by finding the optimal c value and comparing several different kernel functions. Four commonly used kernel functions are linear, polynomial, radial basis function, and sigmoid. Each The best value for each kernel parameter is c = 1.5 for the Linear kernel, c = 15 and gamma 'scale' for Polynomial, c = 5 and gamma 'auto' for Radial Basis Function, and c = 3 and gamma 'auto' for Sigmoid. The comparison results, based on their macro averages are stated below.

TABLE III. SVM MODEL

Kernel Function	Precision	Recall	Accuracy
Linear	0.98	0.97	0.98
Polynomial	0.98	0.97	0.98
Radial Basis Function	0.99	0.98	0.98
Sigmoid	0.96	0.93	0.94

The next method is training the data using MLP. Note that data for the Logistic activation function is not stated as the results for both test data and training data were very low, and as such not used. The results for comparison, based on tuning parameters in the MLP Classifier method are stated below.

TABLE IV. MLP CLASSIFIER MODEL

Activation Function	Hidden Layer	Precision	Recall	Accuracy
ReLu	62, 30	0.90	0.94	0.93
Identity	61	0.97	0.93	0.97
Tanh	61	0.95	0.87	0.93

The comparison of SVM and MLP Classifier model was carry out to get the best model. Table V is the result of the comparison between SVM and MLP Classifier. The accuracy, precision, and recall scores in Table VI demonstrate the high prediction capabilities of both the SVM and MLP Classifier. It can be concluded in this case that SVM with the radial basis kernel is the optimal approach by comparing the accuracy, recall, and precision numbers included in Table VI since this approach yields the best accuracy, precision, and recall numbers when compared to other methods.

TABLE V. COMPARISON OF SVM AND MLP CLASSIFIER

Method	Precision	Recall	Accuracy
SVM	0.99	0.98	0.98
MLP Classifier	0.97	0.93	0.97

VIII. CONCLUSION

By carrying out data training experiments on each method and using different kernels, the reseach was able to give an approximate case on the performances of using both support vector machine and multi-layer perceptron classifiers to classify specific traits in heart disease patients to each level of the disease, which in turn can be a reference for future patient diagnosis. Each method used has shown great numbers and is an indicator that any of the methods might in fact be used for disease predictions in general, not just for heart disease. In this case in particular, the research finds that using SVM classifier with the radial basis function kernel has a higher accuracy than other methods.

It is advised to conduct further experimentations with different datasets to further understand the differences in using each of the methods used in the study against datasets with different traits. The parameter tuning is done mostly manually and might be prone to missing smaller parameters with better statistics and as such a more algorithmic solution is advised. Better methods in data pre-processing is also advised as to find out deeper connection between missing values in data and prediction accuracy.

In the case of heart disease classification, there is an imbalance in the amount of data in each class. Further research should apply techniques such as SMOTE, class weighting, or cost-sensitive learning to improve model performance, especially in minority classes. In addition, in the modeling analysis to obtain the best model, the combination of Support Vector Machine (SVM) and Multilayer Perceptron (MLP) shows potential in improving classification accuracy compared to the single use of each method. Further researchers are advised to explore hybrid model architectures.

Data Availability Statement

The data used in this study are not publicly available due to ethical, legal, and commercial restrictions. The dataset consists of historical patient data related to cancer conditions from a hospital in Indonesia. As such, the data cannot be made available for public access due to privacy concerns and institutional policies regarding the protection of sensitive medical information.

Credit Authorship Contribution Statement

Syarifah Diana Permai: Conceptualization, Supervision, Methodology, Writing - Review & Editing. **Austin Ordell Salomo**: Methodology, Formal analysis. **Fauzan Hafizh Zulfikar**: Data Curation, Visualization. **Muhammad Farras Fadhilah**: Writing - Original Draft. **Ryan Bukhari**: Methodology, Formal analysis. **Muhammad Fadlan Hidayat**: Writing - Review & Editing, Supervision.

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